

Crowe Sense

Prepared 2026-09-01
For a list of 30 investors and funders
Every figure measured that day or marked as a proposal

6	105	\$278	43
surfaces reading one contract: desktop, web, phone, Cortex, terminal, any screen	tests across firmware, relay, CLI, desktop and Cortex integrations, all green	parts per node at 4 GB, priced live 2026-08-31	condition bands from the founder's 68-video corpus, each cited to its clip
0	\$175 _k		
nodes that have run with physical sensors in a room. That is the gap this raise closes	the ask, on a SAFE; cloud and analytics already covered by credits		

Sources: firmware 51, relay 7, CLI 25, desktop 14, Cortex 8 tests; hardware/SHOPPING-LIST.md; analysis/practice-envelope; docs/investor-report.md in github.com/MichaelCrowe11/crowe-sense, branch integrate/2026-09-01 (shared on request until pushed).

IN ONE PARAGRAPH

Crowe Sense is a sensing node for a growing room: a Raspberry Pi 5, four Sensirion and Bosch sensors under a louvered radiation shield, a 3D-printed housing, about \$280 in parts. It keeps every reading on the node, serves them on the local network, and pushes signed batches to a relay so the same numbers appear on six surfaces the

company already ships. Readings are judged against 43 condition bands extracted from the founder's own cultivation corpus, each cited to the clip it came from.

The software is built and tested. No node has yet run with physical sensors in a room. This raise puts twenty-five of them in ten growers' rooms and proves the data with the same test that caught the company's own synthetic stream in August.

The node

Pi 5 • SCD41 • SHT45 • BME688 • VEML7700
SQLite on the node • local API :8078
Ed25519-signed batches out



The relay

one Cloudflare Worker
D1 hot 30 days • R2 raw forever
Crowe ID on every read



Six surfaces, one contract

- desktop
- web
- phone + Mac
- Cortex
- terminal
- House / kiosk

contracts/telemetry-v1.md. The local API and the relay serve identical paths, so a surface does not know or care which one it is reading.

WHAT EXISTS TODAY

Measured on 2026-09-01

PIECE	STATE	EVIDENCE
Firmware 0.2.0: sampler, signed uploader, local API, watchdog, six drivers	built	51 tests, ruff clean
Relay: signed ingest, D1, R2, Crowe ID reads	deployed	sense.crowelogic.com, /health live 2026-09-02; 7 end-to-end tests including signature verification
Contract every surface shares	written	contracts/telemetry-v1.md, JSON schema, D1 schema

Desktop app: the Environment lane fills itself with measured rows	branch	14 tests + 67 parity tests
Terminal: crowe sense, seven subcommands	branch	25 tests; full suite 2,215 green
Cortex panel	branch	8 tests + 28 adjacent; typecheck and build clean
Web edge, /app/sense/*	patched, awaiting deploy	syntax-checked; one-line deploy
Phone and Mac client 0.2.0, direct or cloud source	compiles	cargo check clean
House: embeddable panel and kiosk autostart	built	one tag mounts the whole surface
Enclosures: indoor radiation-shield head; field variant with drive and hotspot bays	parametric OpenSCAD	print settings and tolerances documented
Practice envelope, provenance test, stage and harvest logs	built	23 regression tests, one per mistake it made
Node bill of materials	priced live	\$278 at 4 GB, \$343 at 8 GB

NOT PROVEN, SAID FIRST

What is not proven

No Crowe Sense node has run with physical sensors in a growing room. An earlier daemon ran the whole software path for 23 days in May and June 2026 and produced 3,596,420 rows across 20 metrics. On 2026-08-04 the founder's own provenance test showed the tent temperature, humidity and light channels were not sensor measurements; no sensors had been installed in those tents. The founder confirmed it the same day and wrote a correction to the published Zenodo record (10.5281/zenodo.20722953). As of this revision (2026-09-02) that correction has not yet been filed as a new version of the record. It is available on request, and until it is filed the zone channels in the public record should be read as unverified.

No revenue on this product. The company's software surfaces have no paying subscribers. The founder's businesses have sold cultivation products and courses; none of that is attributed here.

Yield prediction is not built. The dataset that would train it, sensor windows joined to harvest outcomes, has zero labelled rows. The tools to log stages and harvests exist; the pilot fills them.

The reason to say this first: the provenance test is the asset. A company that caught its own synthetic data, published the method, and gates every future dataset on it is selling something a sensor vendor cannot: numbers a grower can trust.

THE FOUNDER

Why this founder

Michael Crowe has grown mushrooms since 2005, starting at fifteen, about ten of those years commercially (Southwest Mushrooms, Phoenix; the farm closed in February 2025). About 342,000 people subscribe to his cultivation channel (YouTube, read 2026-09-02); 68 of those videos were parsed into 642 numeric constants, of which 43 survive as condition bands a sensor can be judged against, with a documented rule that a band needs two independent statements before it judges anything. Nobody else can ship that layer, because nobody else has the corpus. He also built every surface in the table above.

THE DEVICE

The device, in one table

Compute	Raspberry Pi 5, 4 GB, active cooler, 64 GB microSD	
Sensors	SCD41 (NDIR CO2, T, RH), SHT45 (reference T, RH), BME688 (VOC, pressure), VEML7700 (lux); SDP810 differential pressure on the flow-hood node	
Housing	3D-printed PETG, louvered radiation-shield head, about 350 g, 14 h print	
On the node	SQLite (WAL), local API on :8078 with a kiosk page, Ed25519-signed uploads, nothing deleted when sent	
Cloud	one Cloudflare Worker: D1 (30 days hot), R2 (raw batches forever), Crowe ID on every read	
Parts cost	\$278 (4 GB) · \$343 (8 GB)	
Proposed price	\$449 per node, relay included with any paid Crowe ID tier; set after the pilot	

THE ASK

\$175,000 on a SAFE

\$175,000 SAFE

Hardware and people only. Cloud, analytics and model inference run on credits the company already holds: Cloudflare \$99,657 to June 2027 (under 1% used), Azure \$20,000, PostHog \$50,000, and \$13,500 in other unclaimed credits. A \$60,000 first tranche (angel or grant) funds the hardware, the engineering contract, the instruments and half the pilot program, and reaches the day-90 milestone.

USE	AMOUNT
Pilot hardware: 25 nodes, printer, reference instruments, spares	\$12,000
Hardware engineering, contract: sensor-head thermal validation, calibration jig, enclosure DFM, pre-compliance	\$35,000
Founder, twelve months	\$60,000
Pilot program: ten growers, installs, travel, stipends	\$15,000
Compliance and safety review, product insurance	\$10,000
Legal: SAFE, IP assignment, provisional filing on the envelope method	\$8,000
Calibration and reference instruments	\$4,000
Contingency (18%)	\$31,000
Total	\$175,000
Cloud, analytics, model inference	\$0

MILESTONES

Dated from close, each with its proof

day 30	Five nodes in real rooms at the founder's site and two pilot growers first week of data passes provenance_check.py (exit 0), published with the run log
day 90	Twenty-five nodes at ten growers; all six surfaces in weekly use

relay shows 25 nodes reporting on 6 of 7 days; PostHog funnel, paired to day-7 retention, above 80%

day 180	Twenty labelled harvests; Western SARE application submitted (due 2026-10-28, \$35,000, non-dilutive) datasets/outcomes.csv row count; SARE confirmation
day 365	One hundred nodes shipped at \$449; the supervised yield model trained on real labels, or a published reason it cannot be yet shipping records; a model card with held-out error, or the negative result

UNIT ECONOMICS

Per node, proposed

Parts	\$278
Print, assembly, test (1.5 h)	\$40
Landed cost	\$318
Price	\$449
Gross margin	29% at pilot volume; parts fall about 20% at 100 units
Cloud cost	\$0 through June 2027 on credit; under \$1 per node-month after

RISKS

Risks, plainly

RISK	WHAT IS DONE ABOUT IT
Pi 5 pricing (the 2026 DRAM shortage doubled 2024 prices)	4 GB board; the firmware runs on a Pi 4 or a Compute Module if it comes to that
Sensor drift, especially NDIR CO2	SHT45 as reference, SCD41 self-calibration on, every reading carries its driver and a quality flag; reference instruments in the budget
A second provenance failure	the test runs on every node's first week and is published with the data

Founder dependence	every surface, the contract and the setup pipeline are documented in the repo; a contractor built the desktop and Cortex integrations from the contract alone
Market size	not claimed here; the pilot measures whether ten growers keep the nodes plugged in

THE THIRTY

Thirty investors and funders

Ten agtech and controlled-environment funds, six hardware and climate seed funds and accelerators, eight Arizona programs and angel groups, six non-dilutive programs. Every URL was verified active in 2025 or 2026 on 2026-09-01; award ceilings are the current published figures. Contact names are deliberately absent: they change, and a wrong one costs more than a blank. Start with HAX, Techstars Future of Food, Canyon Angels, Desert Angels, Arizona Tech Investors and the USDA and NSF SBIR windows; the large agtech funds are later-round relationships.

Agtech and controlled-environment funds 10

NAME	STAGE OR AWARD	WHY IT FITS
AgFunder	Seed to Series B, broad agrifood and climate tech	AgFunder tracks agrifood infrastructure deals and has backed sensor and data plays comparable to a Raspberry Pi based environmental node for indoor mushroom and CEA growers.
Anterra Capital	\$1,000,000 to \$10,000,000 initial check (stated on the fund's own site)	Anterra's thesis centers on digital and data infrastructure across the agrifood value chain, matching the sensor-plus-Cloudflare-relay architecture behind Crowe Sense.
Astanor Ventures	Seed through growth	Astanor explicitly funds controlled environment agriculture and impact-native agrifood founders, a direct thesis match for an indoor-growing sensing platform.
iSelect Fund	Seed, Series A, crop health and sensing	iSelect's stated mission of funding sensor-based farm management tools and its indoor-farming portfolio company Artemis show current interest in exactly the CO2/VOC/climate sensing hardware Crowe Sense builds.
Ospraie Ag Science	Seed through growth, controlled environment agriculture named as a backed theme	Ospraie explicitly names controlled environment agriculture as a backed theme, making Crowe Sense's CEA sensing node a direct thesis match.

Middleland Capital	Seed, Series A (recent language skews toward de-risked, early-revenue companies)	Middleland's portfolio already includes precision-ag sensing hardware (EarthOptics), showing comfort underwriting physical agricultural sensing plays like Crowe Sense.
Acre Venture Partners	Seed through Series B	Acre's current fund already holds an indoor-farming software company (Source) and ag-robotics hardware bets, placing a hardware-plus-software CEA sensing platform inside their thesis.
Better Food Ventures	Seed, Series A, digital/software-first (deliberately avoids hardware-intensive deals)	Better Food Ventures backs the software side of food and ag systems, so a pitch should lead with the Cloudflare relay, apps and CLI rather than the physical sensor hardware.
Radicle Growth	verify	Radicle's challenge-based sourcing model (2025 AgRogue Growth Partners with Land O'Lakes) is built to surface sensor-and-data farm technology like Crowe Sense.
SVG Ventures THRIVE	Up to \$1,000,000 for Global Impact Challenge finalists (stated by the fund)	THRIVE's recent portfolio includes storage-monitoring sensor company Cellar Insights (Jan 2026), and its Global Impact Challenge funds early agrifood hardware and software at Crowe Sense's exact stage.

Hardware and climate seed funds and accelerators 6

NAME	STAGE OR AWARD	WHY IT FITS
Lemnos	Pre-seed and seed, hardware and deep tech only	Lemnos is a hardware-exclusive early-stage fund with a track record in IoT sensing hardware, matching a Raspberry Pi based environmental node moving from firmware-complete into pilot manufacturing.
HAX (SOSV)	Up to \$550,000 pre-seed; first check \$250,000 (\$150,000 cash plus \$100,000 in-kind engineering and lab support) on an uncapped post-money SAFE (stated on HAX's own FAQ)	HAX is a pre-seed program with in-house mechanical/electrical engineers and small-batch manufacturing support, directly addressing the enclosure, sensor integration and pilot production run Crowe Sense needs.
Lowercarbon Capital	Pre-seed through growth, climate tech with a hardware-heavy sub-portfolio	Lowercarbon funds deep-tech climate hardware and has in-house engineering diligence capacity that could evaluate a CO2/VOC/differential-pressure sensing node as CEA climate infrastructure.
Techstars Future of Food Powered by Ecolab	\$220,000 (Techstars standard terms: \$20,000 for 5% common plus \$200,000 uncapped MFN SAFE)	Successor to Techstars Farm to Fork with an active 2026 cohort; the Cloudflare relay and multi-platform apps behind Crowe Sense fit its data/traceability angle better than the sensor hardware alone.

Y Combinator	\$500,000 (\$125,000 for 7% plus \$375,000 uncapped MFN SAFE)	YC's 2026 Requests for Startups name agriculture robots and physical hardware as priority categories, and its flat \$500k check would fund a Crowe Sense pilot production run outright.
Elemental Impact	First-of-a-kind and early commercial deployment funding across energy, industry, food and agriculture, water	Elemental Impact funds food-and-agriculture climate hardware moving from prototype into real-world deployment, matching Crowe Sense's move from unit-tested firmware into a validated pilot run.

Arizona 8

NAME	STAGE OR AWARD	WHY IT FITS
Desert Angels	Seed/early-stage, accredited angel network	Arizona's top-ranked angel network per the HALO Report gives a Phoenix hardware startup access to the state's deepest bench of accredited investors and sidecar funds.
Canyon Angels	Pre-seed/seed, accredited angel group	A Phoenix angel group meeting on the Grand Canyon University campus is a realistic first local stop for a Phoenix-based hardware founder with an existing regional audience.
Invest Southwest (Venture Madness)	not applicable (event, not a fund)	Arizona's longest-running venture conference puts a Phoenix hardware/agtech founder directly in front of the state's active accredited investors during Arizona Tech Week.
Tallwave Capital	Pre-seed, seed and post-seed VC fund	Named Arizona's most active venture capital firm two years running, Tallwave gives a Phoenix hardware startup a credible local lead investor plus a structured validation track through its High Tide program.
Arizona Tech Investors (ATI)	verify	Arizona's largest angel organization lists advanced manufacturing and sustainability among its target sectors, a direct route to reach 100+ accredited local angels.
PHX Ventures	\$50,000 to \$250,000 in pre-revenue companies (Launch Fund, stated on the fund's own site)	The Launch Fund's stated stage and check-size range matches Crowe Logic almost exactly, though the fund states it invests exclusively in B2B software and AI, not hardware.
AZ TechCelerator	not applicable (space and mentorship, not capital)	A hardware-oriented Arizona incubator offers affordable space and mentors useful for pushing a sensor enclosure and pilot run through validation, though it is not a source of investment capital.

ASU Venture Devils	\$1,000 to \$50,000 per team; Spring 2026 Demo Day pool exceeded \$300,000 in aggregate (ASU's own site)	ASU's venture competition offers non-dilutive prize money at Crowe Logic's exact stage, but eligibility requires an enrolled ASU or Maricopa Community College student as the team's lead founder.
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Non-dilutive programs 6

NAME	STAGE OR AWARD	WHY IT FITS
USDA SBIR/STTR Phase I (NIFA)	\$175,000 maximum for most topics (\$125,000 for topics 8.6 and 8.12), award ceiling not guaranteed	Funds exactly the CO2/VOC/climate sensor hardware and firmware R&D Crowe Sense needs to validate for indoor mushroom and CEA cultivation.
NSF SBIR Phase I (America's Seed Fund)	\$305,000 maximum (solicitation NSF 26-510), award ceiling not guaranteed	NSF explicitly funds small-business deep-tech hardware R&D, and a multi-sensor Pi-based node with a Cloudflare relay is exactly that kind of hardware-plus-software integration project.
Western SARE Farmer/Rancher Grant	\$35,000 flat award	Michael Crowe, as an active mushroom producer, can serve as producer-PI piloting Crowe Sense units on his own grow as a funded on-farm research and outreach project.
USDA NIFA AFRI, Agriculture Systems and Technology	verify (no per-track cap published; check the current NOFO)	This AFRI priority area explicitly covers precision-agriculture sensing and engineering, matching Crowe Sense's environmental sensing platform for CEA more closely than AFRI's larger systems-level tracks.
Arizona Innovation Challenge (Arizona Commerce Authority)	Minimum \$50,000 non-dilutive per awardee, up to 10 awardees per cycle (official ACA July 2025 release)	A state-run non-dilutive competition whose eligible industries include hardware and advanced manufacturing fits a pre-revenue Phoenix hardware company seeking runway toward a pilot run without giving up equity.
Arizona Specialty Crop Block Grant Program (AZDA / USDA AMS)	verify (approximately \$1,250,000 total FY2026 Arizona allocation across all awardees, no per-project cap published)	Cultivated mushrooms are a USDA-recognized specialty crop, so a sensing project framed as a broad benefit to Arizona mushroom and CEA growers, not a single company's product, can qualify.

Crowe Logic, Inc. · github.com/MichaelCrowe11/crowe-sense (public): firmware, hardware and enclosures, docs. Branch `integrate/2026-09-01`, shared on request until it is pushed: relay, contract, archive with the correction, analysis, the six integrations, the shopping list, the setup pipeline, and this report as `docs/investor-report.md`. Prepared 2026-09-01, revised 2026-09-02. Nothing here is a projection dressed as a fact.